DSC 102 — Spring 2026

Midterm Review — Set B

Problem 1

A program stores temperatures in an 8-bit signed integer (two's complement).

Questions:

1. What is the maximum positive value representable?
2. The current reading is 120. A delta of `+15` is added. What decimal value is actually stored?
3. Name the phenomenon that occurred and describe one real-world consequence in a safety-critical system.

Solution 1

1. $2^7 - 1 = \mathbf{127}$
2. $120 + 15 = 135$, and `10000111` interpreted as signed is $-\mathbf{121}$.
3. The phenomenon is **signed integer overflow**. A real sensor could wrap around and report a safe-looking negative value instead of an overheating condition.

Problem 2

Consider a custom 6-bit float:

- Bit 5 = sign s
- Bits 4-3 = exponent
- Bits 2-0 = fraction
- Value $= (-1)^s \cdot 2^{(\exp-1)} \cdot (1 + f_2 2^{-1} + f_1 2^{-2} + f_0 2^{-3})$

What is the decimal value of the bit sequence `010110`?

Solution 2

`0 10 110` gives:

- $s = 0$ so the number is positive
- exponent `10` is 2, so the scale factor is $2^{2-1} = 2$
- fraction `110` gives $1 + 0.5 + 0.25 = 1.75$

So the value is $2 \cdot 1.75 = \mathbf{3.5}$.

Problem 3

Assume:

- 8-bit virtual address
- Page size = 16 bytes
- VPN $\longrightarrow$ PFN mapping: `0→5`, `1→2`, `2→7`, `3→1`

Questions:

1. Translate virtual address 35 to a physical address.
2. Translate virtual address 50 to a physical address.

Solution 3

1. 35: VPN $= 2$, offset $= 3$, PFN $= 7$, so physical address $= 7 \times 16 + 3 = \mathbf{115}$
2. 50: VPN $= 3$, offset $= 2$, PFN $= 1$, so physical address $= 1 \times 16 + 2 = \mathbf{18}$

Problem 4

A program has:

- 25% serial work
- 75% perfectly parallel work
- Run on 8 cores

Questions:

1. What is the maximum theoretical speedup with infinite cores?
2. What is the speedup on 8 cores?
3. Is the speedup linear?

Solution 4

1. $1/0.25 = \mathbf{4}\times$
2. $$\frac{1}{0.25 + 0.75/8} = \mathbf{2.91}\times$$
3. No. The speedup is **sublinear** because of the serial fraction.

Problem 5

A system has:

- L1 access = 4 ns, hit rate = 0.8
- L2 access = 20 ns, hit rate = 0.75 given L1 miss
- DRAM access = 200 ns

Questions:

1. Compute the overall AMAT.
2. What fraction of accesses go to DRAM?
3. If L1 hit rate improves to 0.9, recompute AMAT.

Solution 5

1. $$0.8(4) + 0.2\left[0.75(20) + 0.25(200)\right] = \mathbf{16.2}\ \mathrm{ns}$$
2. $0.2 \times 0.25 = \mathbf{5\%}$
3. $$0.9(4) + 0.1\left[0.75(20) + 0.25(200)\right] = \mathbf{10.1}\ \mathrm{ns}$$

Problem 6

A programmer writes two loops to sum a `1000x1000` integer matrix stored in row-major order:

- Version A: `for i in rows: for j in cols: total += M[i][j]`
- Version B: `for j in cols: for i in rows: total += M[i][j]`

Questions:

1. Which version runs faster on a modern CPU? Explain why in terms of cache lines and spatial locality.
2. What cache behavior does the slower version exhibit?

Solution 6

1. **Version A** runs faster because it walks through contiguous memory and uses spatial locality well.
2. **Version B** exhibits **cache thrashing** because it jumps through memory with a large stride and tends to miss on each access.

Problem 7

A 12 GB row-oriented file has 120 equal-size columns.
A query only needs one column and takes 30 seconds.

Questions:

1. How much useful data is needed?
2. What is the effective useful throughput?
3. If SSD bandwidth is 600 MB/s, what percentage is useful throughput?

Solution 7

1. $12\ \mathrm{GB}/120 = 0.1\ \mathrm{GB} = \mathbf{100}\ \mathrm{MB}$
2. $100/30 = \mathbf{3.33}\ \mathrm{MB/s}$
3. $\dfrac{3.33}{600} \approx \mathbf{0.56\%}$

Problem 8

A program executes 2 billion memory accesses.

- Cache hit time = 4 cycles
- Miss penalty = 80 cycles
- Hit rate = 0.85
- CPU clock = 2 GHz

Questions:

1. Compute average cycles per memory access.
2. Compute total execution time.
3. If hit rate improves to 0.95, what is the new execution time?
4. What is the speedup?

Solution 8

1. $4 + 0.15(80) = \mathbf{16}$ cycles/access
2. $\dfrac{2 \times 10^9 \cdot 16}{2 \times 10^9} = \mathbf{16}\ \mathrm{s}$
3. New average $= 4 + 0.05(80) = 8$, so the new execution time is $\mathbf{8}\ \mathrm{s}$.
4. $\dfrac{16}{8} = \mathbf{2}\times$

Problem 9

A system uses 32-bit virtual addresses and a page size of 4 KB.

Questions:

1. How many bits are used for the page offset?
2. How many bits are used for the virtual page number?
3. How many entries are in a flat single-level page table for one process?

Solution 9

1. $4\ \mathrm{KB} = 4096 = 2^{12}$, so the page offset uses $\mathbf{12}$ bits.
2. $32 - 12 = \mathbf{20}$ bits for the virtual page number.
3. $2^{20} = \mathbf{1{,}048{,}576}$ page-table entries

Problem 10

A multi-threaded analytics job runs on a 2-socket NUMA server.

- Remote RAM access takes 3x longer than local RAM access.
- Local RAM latency = 100 ns.
- Thread A (on Socket 0) reads a 4 GB array stored in Socket 1's RAM.
- Thread B (on Socket 1) reads a 4 GB array stored in Socket 1's RAM.

Questions:

1. What effective latency does Thread A experience per memory access?
2. Both threads write to adjacent 64-bit integers that share the same 64-byte cache line. What performance problem arises?
3. A data engineer proposes copying Thread A's 4 GB array to Socket 0's RAM before the job starts. What is the trade-off?

Solution 10

1. $3 \times 100 = \mathbf{300}\ \mathrm{ns}$
2. The problem is **false sharing**.
3. The trade-off is faster local access for Thread A, but it costs an extra 4 GB of RAM plus a one-time copy before computation starts.

Problem 11

A single-core system runs four processes:

Process	Burst
P1	20 ms
P2	5 ms
P3	15 ms
P4	10 ms

All arrive at time 0. Context switch overhead is 1 ms per switch, charged before each process except the first.

Questions:

1. Under FIFO scheduling (`P1, P2, P3, P4`), what is the average waiting time?
2. Under SJF (non-preemptive), what is the average waiting time?
3. How many milliseconds total are consumed by context switches under each algorithm?

Solution 11

1. FIFO start times are `0`, `21`, `27`, `43`, so the waiting times are `0`, `21`, `27`, `43`.
   Average waiting time $= 91/4 = \mathbf{22.75}\ \mathrm{ms}$
2. SJF order is `P2`, `P4`, `P3`, `P1`, with start times `0`, `6`, `17`, `33`.
   Average waiting time $= (0 + 6 + 17 + 33)/4 = \mathbf{14}\ \mathrm{ms}$
3. There are 3 switches in either schedule, so the switch cost is $\mathbf{3}\ \mathrm{ms}$.

Problem 12

Four jobs arrive at $t = 0$:

Job	Burst
P1	9
P2	5
P3	3
P4	7

Round Robin quantum = 3.

Questions:

1. What is the completion time of each process?
2. What is the average turnaround time?
3. What is the average response time?

Solution 12

Schedule:

`P1 0–3`, `P2 3–6`, `P3 6–9`, `P4 9–12`, `P1 12–15`, `P2 15–17`, `P4 17–20`, `P1 20–23`, `P4 23–24`

1. Completion times: $\mathbf{P1 = 23,\ P2 = 17,\ P3 = 9,\ P4 = 24}$
2. Average turnaround time $= (23 + 17 + 9 + 24)/4 = \mathbf{18.25}$
3. Average response time $= (0 + 3 + 6 + 9)/4 = \mathbf{4.5}$